# Yu-Han Huang

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## EDUCATION

**B.S. in Electrical Engineering, National Taiwan University (NTU)**, Taipei, Taiwan — Sep 2023–June 2027 (expected)

- Overall GPA: 4.04/4.30 (3.91/4.0 converted)
- Selected Coursework: Machine Learning, Deep Learning for Computer Vision, Deep Reinforcement Learning
- Research Interests: Speech and Audio Processing, GUI Agents, Vision-Language-Action Models

## RESEARCH & INTERNSHIP EXPERIENCES

**Synology Inc.**, New Taipei City, Taiwan — Nov 2026–Present

Video Deep Learning Product Developer Intern

**Natural Language Processing Lab (NLP), UCLA** — July 2026–Present

Visiting Researcher | Supervisor: Prof. Kai-Wei Chang

- Investigated GUI agents for multi-step web navigation and computer-use tasks
- Developed realistic tasks that required GUI agents to combine deep research with multi-step browser interactions

**Speech Processing and Machine Learning Lab, NTU** — July 2025–Present

Undergraduate Researcher | Advisor: Prof. Hung-Yi Lee

- Explored training-free approaches to improve audio model understanding and response reliability
- Investigated safety and alignment methods for full-duplex spoken language models

**Open Cloud Infrastructure Software Center (OCIS), ASUS** — May 2026–Nov 2026

AI Research Intern

- Conducted research on capability enhancement, evaluation, safety, and alignment for large audio-language models, full-duplex speech models, and LLM-based agents

**Vision and Learning Lab, NTU** — Sep 2025–June 2026

Undergraduate Researcher | Advisor: Prof. Yu-Chiang Frank Wang

- Studied VLA models for long-horizon embodied task planning

**Institute of Information Science, Academia Sinica** — July 2024–Aug 2024

Student Researcher | Supervisor: Prof. Yennun Huang

- Developed automated pipelines for generating both attack and benign audit logs for intrusion detection research

## PUBLICATIONS

[*] *Equal contribution*

**[1]** **Yu-Han Huang**, Chih-Kai Yang[*], Ke-Han Lu[*], Cheng An Yu[*], and Hung-yi Lee, "Enhancing How Large Audio-Language Models Hear: Identifying and Amplifying Acoustic Neurons in the Audio Encoder." **[arXiv:2607.11801]**. Accepted to the *2026 IEEE Spoken Language Technology Workshop (SLT)*.

**[2]** Kai-Po Chang, Chang-Hsun Wu, Yen-Siang Wu, **Yu-Han Huang**, Kai-Chen Yang, and Yu-Chiang Frank Wang, "Self-Correcting Imaginative Planning for VLM Embodied Agents," submitted to the *International Conference on Learning Representations (ICLR 2027)* (under review).

**[3]** Lok-Lam Ieong[*], Chia-Chien Chen[*], Chih-Kai Yang[*], **Yu-Han Huang**, An-Yu Cheng, and Hung-yi Lee, "Nudging Hidden States: Training-Free Model Steering for Chain-of-Thought Reasoning in Large Audio-Language Models," in *Proc. Interspeech 2026*, 2026, pp. 3797–3802, doi: 10.21437/Interspeech.2026-554.

**[4]** Kai-Wei Chang, Yi-Cheng Lin, Huang-Cheng Chou, Wenze Ren, **Yu-Han Huang**, Yun-Shao Tsai, Chien-Cheng Chen, Yu Tsao, Yuan-Fu Liao, Shrikanth Narayanan, James Glass, and Hung-yi Lee, "TaigiSpeech: A Low-Resource Real-World Speech Intent Dataset with Scalable Data Mining In-the-Wild," in *Proc. Interspeech 2026* (Long Paper Track), 2026, pp. 7304–7313, doi: 10.21437/Interspeech.2026-1511.

**[5]** Ke-Han Lu et al., including **Yu-Han Huang**, "How Auditory Knowledge in LLM Backbones Shapes Audio Language Models: A Holistic Evaluation." **[arXiv:2603.19195]**. Accepted to the *2026 IEEE Spoken Language Technology Workshop (SLT)*.

**[6]** Yi-Cheng Lin et al., including **Yu-Han Huang**, "Findings of the First Teaching Monster Challenge: A Benchmark of Pedagogical Content Knowledge in AI Agents." **[arXiv:2608.08852]**, submitted to the *AAAI 2027 Artificial Intelligence for Social Impact Track* (under review).

## TEACHING EXPERIENCES

**Deep Learning for Human Language Processing (CommE5054)**, EECS Department, NTU — Spring 2026

Teaching Assistant | Instructor: Prof. Hung-Yi Lee

- Co-organized the First Teaching Monster Challenge and co-developed its baseline pipeline
- Conducted human evaluation of AI-generated teaching videos and coauthored the paper

**Machine Learning (EE5184)**, EECS Department, NTU — Spring 2026

Teaching Assistant | Instructor: Prof. Hung-Yi Lee

- Designed and prepared the course assignment on model merging

## AWARDS & SCHOLARSHIPS

**AI Talent+ Overseas Research Scholarship**, NTU — July 2026–Aug 2026

- Natural Language Processing Lab, UCLA (Prof. Kai-Wei Chang)

**MakeNTU Hackathon – 1st Place, Enterprise Award**, Taipei, Taiwan — May 2025

- Project: **AI Driving Coach**
- Integrated WebSocket streaming and the Google Gemini API for driver analysis, voice feedback, and data visualization
- Built an integrated hardware-software architecture to simulate real-world driving scenarios

## COURSE PROJECTS

**Drifting Q-Learning with Distributional Behavior Cloning** — Spring 2026

*Deep Reinforcement Learning*

- Co-developed DriftQL, an offline reinforcement learning method that replaces pointwise behavior cloning with a distributional drifting objective
- Conducted experiments against 13 baselines on 10 OGBench tasks, achieving state-of-the-art performance on three tasks with one-step inference
- Oral presentation, First National Taiwan University Deep Reinforcement Learning Course Mini-Conference 2026

**Detection-Guided Image Layer Extraction** — Fall 2025

*Deep Learning for Computer Vision*

- Built a two-stage image-decomposition pipeline using Qwen-Image-Layered for RGBA layer generation
- Refined layer boundaries with DocLayout-YOLO-based detection, reducing cross-layer content leakage

## LEADERSHIP & EXTRACURRICULAR ACTIVITIES

**General Coordinator**, NTUEE Night, NTU — March 2026

- Led the department's largest annual event with 200+ performers and staff, attracting an audience of 800+

**Director of Public Relations**, NTUEE Student Association — Fall 2025–Spring 2026

- Led the Public Relations Department and managed vendor outreach and partnerships for student association events

**Vice Captain**, NTUEE Women's Basketball Team — Fall 2024

## SKILLS

**Programming & frameworks:** Python, C/C++, Go, PyTorch, RISC-V, MATLAB

**Tools & platforms:** Git, Docker, Raspberry Pi, HTML/CSS

**Languages:** Mandarin Chinese (native), English (TOEFL iBT 5.5/6, 110/120)